EE3621: Digital Signal Processing

Lecture 10: FIR Filter Design — Window Method
(III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins):** Introduction to FIR vs IIR trade-offs and the concept of linear phase.
- **05:00 – 12:00 (7 mins):** Linear phase conditions for FIR filters and group delay.
- **12:00 – 20:00 (8 mins):** Ideal frequency responses and derivation of the ideal impulse response (LPF).
- **20:00 – 25:00 (5 mins):** The problem of truncation and introduction of the Window Method.
- **25:00 – 32:00 (7 mins):** Common window types and their specifications.
- **32:00 – 36:00 (4 mins):** Step-by-step design procedure.
- **36:00 – 40:00 (4 mins):** Worked Example: 21-tap Hamming-windowed LPF and Checkpoint Questions.

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2 FIR vs IIR Trade-offs

When designing digital filters, we primarily choose between Infinite Impulse Response (IIR) and Finite Impulse Response (FIR) filters.

2.1 FIR Filter Advantages

- **Strictly Linear Phase:** FIR filters can be designed to have exact linear phase. Linear phase means that all frequency components experience the same time delay (group delay), preventing phase distortion.
- **Inherent Stability:** Because the impulse response is finite, there is no feedback loop. The poles of the transfer function are always located at the origin ($z = 0$), guaranteeing bounded-input bounded-output (BIBO) stability.
- **Implementation Simplicity:** FIR filters can be implemented efficiently using fast convolution (FFT) or specialized DSP hardware.

2.2 FIR Filter Disadvantages

- **Higher Order Requirement:** To achieve the same magnitude specifications as an IIR filter, an FIR filter typically requires a much higher order, meaning more memory and computations.

3 Linear Phase Condition in FIR Filters

An FIR filter has a transfer function:

$$H(z) = \sum_{n=0}^{M-1} h[n]z^{-n} \tag{1}$$

where M is the filter length.

3.1 Symmetry and Antisymmetry

A causal FIR filter possesses exact linear phase if its impulse response $h[n]$ satisfies:

- **Symmetric:** $h[n] = h[M-1-n]$
- **Antisymmetric:** $h[n] = -h[M-1-n]$

The group delay is $\alpha = \frac{M-1}{2}$. There are four types of linear phase FIR filters depending on symmetry and whether M is even or odd.

4 Ideal Frequency Responses

The four basic ideal filter types defined over $-\pi \leq \omega \leq \pi$:

- **Ideal LPF:** $H_d(e^{j\omega}) = 1$ for $|\omega| \leq \omega_c$, 0 otherwise.
- **Ideal HPF:** $H_d(e^{j\omega}) = 0$ for $|\omega| < \omega_c$, 1 otherwise.
- **Ideal BPF:** $H_d(e^{j\omega}) = 1$ for $\omega_{c1} \leq |\omega| \leq \omega_{c2}$, 0 otherwise.
- **Ideal BSF:** $H_d(e^{j\omega}) = 0$ for $\omega_{c1} < |\omega| < \omega_{c2}$, 1 otherwise.

5 Ideal Impulse Response (Ideal LPF)

Using the Inverse Discrete-Time Fourier Transform (IDTFT):

$$h_d[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(e^{j\omega})e^{j\omega n}d\omega = \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{j\omega n}d\omega \tag{2}$$

$$h_d[n] = \frac{1}{2\pi jn}\left(e^{j\omega_c n} - e^{-j\omega_c n}\right) = \frac{\sin(\omega_c n)}{\pi n} \tag{3}$$

For $n = 0$, $h_d[0] = \frac{\omega_c}{\pi}$. The response extends to $\pm\infty$, making it non-causal and unrealizable.

6 Truncation and Windowing Method

We truncate $h_d[n]$ and shift it to ensure causality. Multiplying the shifted $h_d[n]$ by a finite window function $w[n]$ suppresses the Gibbs phenomenon.

$$h[n] = h_d\left[n - \frac{M-1}{2}\right] \cdot w[n], \quad 0 \leq n \leq M-1 \tag{4}$$

7 Window Types with Specifications

Let $N = M - 1$.

- **Rectangular:** $w[n] = 1$. Mainlobe width $\frac{4\pi}{M}$, Peak Sidelobe -13 dB.
- **Hanning:** $w[n] = 0.5 - 0.5\cos\left(\frac{2\pi n}{M-1}\right)$. Mainlobe width $\frac{8\pi}{M}$, Peak Sidelobe -31 dB.
- **Hamming:** $w[n] = 0.54 - 0.46\cos\left(\frac{2\pi n}{M-1}\right)$. Mainlobe width $\frac{8\pi}{M}$, Peak Sidelobe -41 dB.
- **Blackman:** $w[n] = 0.42 - 0.5\cos\left(\frac{2\pi n}{M-1}\right) + 0.08\cos\left(\frac{4\pi n}{M-1}\right)$. Mainlobe width $\frac{12\pi}{M}$, Peak Sidelobe -57 dB.
- **Kaiser:** Parameterized by β allowing continuous trade-off.

$$w[n] = \frac{I_0\left(\beta\sqrt{1 - \left(\frac{n-\alpha}{\alpha}\right)^2}\right)}{I_0(\beta)} \tag{5}$$

8 Design Procedure Step-by-Step

1. Determine specifications $(\omega_p, \omega_s, A_s)$.
2. Select the appropriate window based on required attenuation A_s.
3. Compute the required filter length M using the chosen window's mainlobe width formula $\frac{C\pi}{M} \leq \omega_s - \omega_p$.
4. Compute the delayed ideal impulse response $h_d[n - \alpha]$.
5. Compute the window function $w[n]$.
6. Multiply to get causal coefficients $h[n] = h_d[n - \alpha]w[n]$.
7. Verify symmetry for linear phase.

9 Worked Example: 21-tap Hamming-windowed LPF

Design a 21-tap causal FIR Lowpass Filter with cutoff frequency $\omega_c = 0.4\pi$ using a Hamming window.

- $M = 21 \implies \alpha = 10$.
- $h_d[n - 10] = \frac{\sin(0.4\pi(n-10))}{\pi(n-10)}$ for $n \neq 10$, and 0.4 for $n = 10$.
- $w[n] = 0.54 - 0.46\cos\left(\frac{2\pi n}{20}\right)$.

Computing the center tap $n = 10$:

$$h[10] = 0.4 \times (0.54 - 0.46\cos(\pi)) = 0.4 \times 1.0 = 0.4 \tag{6}$$

Computing $n = 9$:

$$h[9] = \frac{\sin(0.4\pi)}{\pi} \times (0.54 - 0.46\cos(0.9\pi)) \approx 0.3027 \times 0.9775 \approx 0.2959 \tag{7}$$

10 Table of Key Formulas

Concept	Formula
Ideal LPF $h_d[n]$	$\frac{\sin(\omega_c n)}{\pi n}$
Group Delay α	$\frac{M-1}{2}$
Causal FIR $h[n]$	$h_d[n-\alpha] \cdot w[n]$
Transition Band $\Delta\omega$	$\omega_s - \omega_p$
Linear Phase Condition	$h[n] = \pm h[M-1-n]$

11 Checkpoint Questions

1. **Why is an ideal Lowpass filter physically unrealizable?**
   Answer: Its impulse response $\frac{\sin(\omega_c n)}{\pi n}$ extends to $-\infty$, making it non-causal. It also extends to $+\infty$, requiring infinite memory.

2. **For 50 dB stopband attenuation and 0.05π transition band, find the window and minimum M.**
   Answer: We need at least Blackman (-57 dB). Setting $\frac{12\pi}{M} \leq 0.05\pi \implies M \geq 240$.

3. **Prove that a symmetric impulse response $h[n] = h[M-1-n]$ has linear phase.**
   Answer: Extracting $e^{-j\omega\alpha}$ from $H(e^{j\omega})$ allows pairing $h[n]$ terms to form real cosine terms, yielding a purely real amplitude function multiplied by a linear phase factor $e^{-j\omega\alpha}$.